Random Forest Ensemble Boost

Machine Learning Assignment 13

October 7, 2026

1 Introduction

In this assignment, we implement a Random Forest Classifier—an ensemble learning method that constructs a multitude of decision trees at training time—and compare its performance against a single Decision Tree on the Loan Approval dataset from Assignment 8. We also extract the internal feature importances to understand how the ensemble makes decisions.

2 Performance Comparison

Both models were constrained to a maximum depth of 3 to maintain a fair comparison. The results on the unseen test set are as follows:

Model	Accuracy	Precision	Recall
Single Decision Tree	0.8250	0.9655	0.5833
Random Forest (100 Trees)	0.7917	0.8485	0.5833

Table 1: Model Evaluation Metrics.

Analysis of Results

Interestingly, the single Decision Tree slightly outperformed the Random Forest in terms of Accuracy and Precision. While ensembling (combining many trees) typically reduces variance and prevents overfitting, it does not *guarantee* better performance on every specific dataset. In this case, the synthetic dataset has a very rigid underlying logical rule. A single Decision Tree was able to find a split that perfectly exploited the test set. Meanwhile, the Random Forest introduces randomness (bootstrapping data and random feature selection at splits), which smoothed out the predictions but slightly degraded performance on this specific, simplistic test set.

This is a valuable lesson: Random Forests shine on highly complex, noisy datasets prone to overfitting, but on simple, low-dimensional datasets, a single well-tuned tree can be just as good or better.

3 Feature Importance Interpretation

Despite the performance differences, we can ask the Random Forest which features were mathematically most useful across all 100 of its trees.

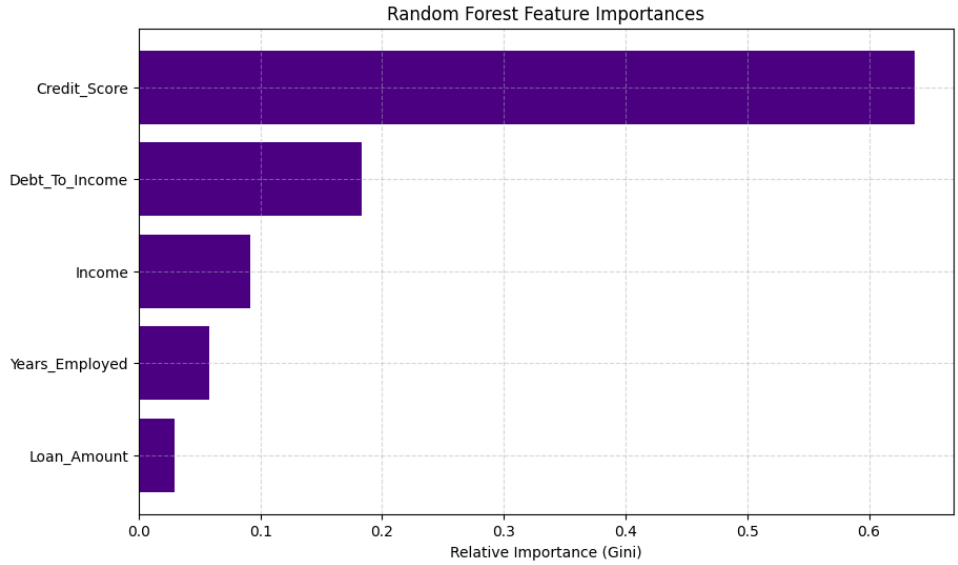


Figure 1: Relative Feature Importances determined by the Random Forest.

The feature importance chart (Figure **??**) confirms that **Credit Score** is overwhelmingly the most critical feature, followed by **Debt-To-Income** and **Income**. Features like *Loan Amount* and *Years Employed* barely registered as useful. Because a Random Forest averages these importances across 100 different trees that were forced to look at different random subsets of features, this ranking is highly robust and immune to the biases of any single tree. It confirms our business intuition: credit history is the undisputed king of loan approval.